

Second Edition

Basic Concepts *of* Environmental Chemistry

Des W. Connell



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Boca Raton London New York Singapore

A CRC title, part of the Taylor & Francis imprint, a member of the Taylor & Francis Group, the academic division of T&F Informa plc.

CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

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Version Date: 20110714

International Standard Book Number-13: 978-0-203-02538-3 (eBook - PDF)

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Preface

In recent years, there has been an extraordinary development in our knowledge of the behavior and effects of chemicals in the environment. In the 1960s, our knowledge of environmental chemistry was confined principally to sets of data on the concentrations of chemicals in the environment. Now a body of specific knowledge has been developed that provides a theoretical basis for understanding distribution, transformation, toxicity, and other environmental properties of chemicals. Out of this has emerged a new branch of the discipline of chemistry that can be defined as follows:

Environmental chemistry is the study of sources, reactions, transport and fate of chemical entities in the air, water and soil environments as well as their effects on human health and the natural environment.

Environmental chemistry is probably the most interdisciplinary of all the branches of chemistry since it includes aspects of all the traditional chemistry branches, as well as biochemistry, toxicology, limnology, ecology, and so on. It has a practical focus on the environmental management of chemicals since that is a major area of concern. It also includes behavior of natural chemical entities in natural systems.

Environmental chemistry now has sufficient intellectual depth to provide the scope for undergraduate and graduate courses in chemistry. This book utilizes the fundamental properties of bonds and molecules as a framework for understanding the behavior and effects of environmental chemicals. The properties of common contaminants are illustrated using environmental behavior, rather than laboratory behavior, as the expression of the underlying bonding and molecular characteristics. It goes on to link these characteristics to biological effects in the environment, such as toxicity. Also, these fundamental aspects are utilized in considering the great global environmental chemistry processes, including respiration, photosynthesis, and so on. Recent developments in the management of hazardous chemicals, such as ecotoxicology and risk assessment, are treated as an aspect of environmental chemistry.

This book assumes only a basic prior knowledge of chemistry and is designed as a textbook for courses in environmental chemistry. Its objective is to provide knowledge of environmental chemistry based on a series of theoretical principles, rather than as a set of disjointed facts. It aims to give students not just knowledge of environmental chemistry, but an understanding of how and why processes in the environment occur. It can also be used as a basis for understanding chemistry, in general, since the principles apply generally as well as to the environment.

We have developed this textbook based on our experience of teaching graduate and undergraduate courses on environmental chemistry over the last 25 years. To

some extent, this represents a culmination of this experience and a distillation of the knowledge we have acquired in that time.

Many people have assisted us in preparing the first and second editions of this book. Greg Miller was involved in discussions of the content and scope, Mrs. Rahesh Garib typed the difficult manuscript, and Chris Clayton and Aubrey Chandica prepared the diagrams. To all of these people, we are extremely grateful.

Des W. Connell

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Part I

Principles of Environmental Chemistry

1 The Role and Importance of Environmental Chemistry

1.1 ALCHEMY

Modern chemistry has its early beginnings in ancient Greece and Egypt. The Greeks were great philosophers and thinkers and debated the ultimate nature of the material world that surrounded them. On the other hand, the Egyptians were skilled in the practical arts of applied chemistry. They had an intimate knowledge of such matters as embalming of the dead, dyeing of clothing, and the isolation of some metals. In fact, the word *alchemy* is derived from the Arab word *al-kimiya*. This phrase was used to describe the art of transformation of materials and the practical use of chemicals in society. However, the lack of understanding of chemical processes led to alchemy being more of an art than a science. The art of alchemy was closely related to religion and involved the use of a set of hieroglyphics to represent the different metals. For example, gold was depicted as the sun, silver as the moon, copper as Venus, and so on. The occurrence of chemical changes was interpreted in a mysterious fashion and took on a mythological significance. The major area of concern of alchemy was not the development of a science, but the transmutation of metals, particularly the transmutation of base metals such as lead into gold.

The decline of alchemy was signaled by the works of such scientists as Agricola, in fact a German named Georg Bauer, who lived between 1494 and 1555. Agricola published a book titled *De Re Metallica* [*Of Metallurgy*], which took a practical approach and gathered together all of the knowledge available on metallurgy at the time. Also, the German scientist Andreas Libau (1540–1616), better known as Libavius, published *Alchemia* in 1597. Despite the name of this text, the book is written clearly and without resort to mysticism. The famous Swiss physician Paracelsus (1493–1541), although an alchemist, believed that the processes that occur in the human body are basically of a chemical nature and that chemical medicines could provide remedies to illness. So this marks the start of the development of chemotherapy, and doctors everywhere now use a vast array of chemicals to cure all kinds of illnesses.

Perhaps the end to alchemy occurred in 1661, when Robert Boyle, an Irishman, published his book *The Sceptical Chymist*. Boyle attacked the ideas and approach of the alchemists and advocated a rational scientific approach. To mark the change from alchemy, Boyle also advocated the use of the term *chemistry* to describe the science of materials. The well-known Boyle's law was developed and amply illustrates the scientific approach to chemistry that he promoted.

It could be said that **environmental chemistry** had its beginning with one of the great pioneers of chemistry, Antoine Lavoisier, who was born in Paris in 1743. Lavoisier's experiments on the atmosphere mark a great advance in chemistry and also a great advance in understanding the chemistry of the atmospheric environment. Not only did he discover fundamental information concerning the chemistry of air, but he also examined the use of air by animals and, by so doing, was investigating one of the major aspects of the chemistry of the environment. The use of oxygen by animals, and their consequent release of carbon dioxide, is a fundamental aspect of environmental chemistry. Lavoisier's *Elementary Book of Chemistry*, published in 1789, effectively marks the start of the systematic development of the then new science of chemistry.

1.2 THE CHEMICAL AGE

In the first half of the 19th century, early chemists were endeavoring to describe the nature of molecules. They were taking the first steps in assembling the structures of organic chemicals, which we now take for granted as common scientific knowledge. As they were carrying out this early work, they were concerned with the reactions between organic chemicals and the nature of the products generated, thus embarking on the path to modern synthetic organic chemistry.

One of these early chemists, William Perkin (1838–1907), became aware that some of these synthetic organic chemical processes could possibly be turned to commercial advantage. Perkin left his academic studies at the university and used money obtained from his family to start a factory to manufacture synthetic dyes. One dye he produced, aniline purple, was extremely popular in the textile industry and in heavy demand with textile manufacturers in Europe. In fact, Perkin had founded the first chemical industry to be based on studies of the nature of organic compounds conducted in a research environment. Not only had he founded the synthetic chemical industry, but he was also able to retire a very wealthy man at an early age. This marks the start of the synthetic chemical industry, which, from this small beginning, was to expand to a size and extent where its products now dominate modern society. This development had important implications for the environment, since it resulted in the preparation and discharge of relatively large quantities of synthetic organic substances, previously unknown to occur in nature and with unknown environmental effects.

Of course, chemical processes such as the smelting of ores and the manufacture of soap had been carried out on a large commercial scale prior to this time. But these processes were developed as a result of trial and error and were more in the nature of trade practices rather than the products of scientific chemical research.

Another chemist whose activities were to have important environmental implications was on the scene about this time: German scientist Fritz Haber (1868–1934). Haber demonstrated in the laboratory how nitrogen and hydrogen could be combined to yield ammonia. He further developed this on a commercial scale so that commercial plants operating in Germany in the early part of this century were fixing nitrogen gas from the atmosphere and producing ammonia. This ammonia could be used as a fertilizer for the production of food crops, thereby eliminating the need to rely on

TABLE 1.1
Use of Synthetic Chemicals in Human Society

Substance	Examples of Chemical Classification	Uses
Plastics	Polystyrene, polyvinylchloride polypropylene, nylon	Textiles, car tires, household goods, furniture, etc.
Pesticides	DDT, 2,4-dichlorophenoxyacetic acid (2,4-D), malathion, glyphosate	Control of weeds, insects, and other pests
Drugs	Aspirin, penicillin, Valium, sulfanilamide, barbiturates	Medicinal uses
Petroleum products	Hydrocarbons	Motor fuel, lubricant
Crop fertilizer	Ammonia, ammonium sulfate, ammonium nitrate, ammonium phosphate	Stimulate food crop production

the natural occurrence of nitrogen compounds in soil to stimulate plant growth. This process has continued to be used to the present day, thereby allowing the production of large quantities of plant fertilizer used to produce food crops and allowing rapid expansion of the world's population. Without this process, the production of food would be limited by the availability of natural nitrogen compounds in soil for food plant growth. Thus, this chemical process has probably had the greatest impact on the environment through the rapid expansion of the human population with the subsequent environmental changes that have resulted.

It is traditional for human history to be divided into a series of ages, each building on and improving the technology of the previous. Thus, we have a sequence of ages going from the Stone Age to the Bronze Age to the Industrial Age and the Atomic Age. However, during the last approximately 100 years, the age in which we are living could very well be described as the **Chemical Age**. Table 1.1 indicates the wide range of synthetic chemicals now used in human society. The use of these substances permeates almost every aspect of our society, from life in the home to transport, food production, availability of medicines, and so on. The production and use of these many substances have improved human conditions and caused an enormous increase in the human population. These factors have many implications for the chemical processes that occur in the environment.

1.3 ENVIRONMENTAL CHEMISTRY

Environmental chemistry is not new, but the use of this term to describe a body of chemical knowledge has become accepted by scientists in recent years. This principally reflects the importance attached to the behavior and possible adverse effects of chemicals discharged to the environment.

There are many scientific events that could now be identified as the starting points of environmental chemistry. Some of these are in the area of fundamental chemistry, while others lie in applied chemistry. Also, the chemical aspects of other disciplines have played a major role. One of the starting points has been mentioned previously: the work of Lavoisier. His analysis of air and determination of the nature

of combustion are fundamental to the development of modern chemistry and atmospheric chemistry. Turning to the applications of chemistry, perhaps pollution of the Thames River during the 1800s can be identified as another starting point of environmental chemistry. The bad odors and diseases associated with the Thames were investigated by various British Royal Commissions, which weighed evidence on the chemistry of sewage treatment and water pollution. This culminated with the Royal Commission on Prevention of River Pollution, which reported in 1885 and recommended the use of the biochemical oxygen demand (BOD) test. This test is still extensively used today to evaluate the effects of sewage and other wastewaters on waterways.

The chemistry of the oceans, lakes, and freshwater areas also has a comparatively long history. In 1872, HMS *Challenger* commenced its historic voyages with many of the most noted men of science at that time aboard. They conducted many investigations of seawater composition and chemical processes in the oceans. Similarly in limnology, the key to understanding the ecology of freshwater areas lies in chemical transformations of carbon, oxygen, nitrogen, and phosphorus. This was recognized early in the development of this discipline, which has always had a strong emphasis on chemistry.

Since these early beginnings, environmental chemistry has expanded rapidly. Natural chemical processes in all sectors of the environment, particularly soil, water, and the atmosphere, have been subject to investigation, as well as the environmental behavior of contaminating chemicals. Not surprisingly, the management of chemicals discharged to the environment has become a major focus for environmental chemistry. Governmental agencies and industries employ large numbers of environmental chemists on a worldwide basis to monitor and manage the discharge of chemicals and their adverse effects.

Over recent years, a body of knowledge associated specifically with the behavior and effects of chemicals in the environment has been developing. A theoretical basis for understanding the distribution, transformation, toxicity, and other biological properties of chemicals in the environment is now becoming established. This means that environmental chemistry can be appropriately seen as a subbranch or subdiscipline of chemistry. This is reflected in the availability of excellent textbooks on the topic as illustrated by those by Manahan (2001) and Schwartzenbach et al. (2002). The recent development of environmental chemistry was initiated during the 1960s. Rachel Carson's book *Silent Spring*, published in 1962, can be identified as a significant event stimulating worldwide interest and concern regarding chemical residues in the environment. This event was made possible by the development of analytical techniques capable of detecting chemicals at very low concentrations. In 1952, Richard Synge and Archer Martin were awarded the Nobel Prize for inventing the chromatography technique, which has a principal application to the analysis of organic compounds. Later, the technique was extended and improved such that trace amounts of xenobiotic organic chemicals could be quantified in environmental samples. The flame ionization detector extended the sensitivity of the method so that levels of a few parts per million of organic compounds could be detected in samples. In addition, with the chlorinated hydrocarbons, the development of the electron capture detector further extended the sensitivity of the technique. In 1967, only 15 years after the Nobel Prize for chromatography was

awarded, a considerable body of information was available on the concentrations of xenobiotic organic compounds in biota.

Environmental chemistry was dominated at this time by the collection of data on residues of synthetic compounds in biota, but there was little understanding of the mechanisms, how the residues accumulated, or their biological effects. Determination of residue levels remains an important aspect of chemical behavior in the environment, but now there is considerable interest in placing residues in a broader context of environmental effects. There have been major advances in understanding the distribution, transport, and transformation of contaminants, as well as exposure and uptake by biota. But human health effects as well as the responses of natural ecosystems require further expansion of knowledge (Boethling and Mackay, 2000; Lyman et al., 1990).

Over the last 15 years, some theoretical concepts have emerged that provide a sound conceptual basis for important aspects of environmental chemistry. Drawing on ideas already established in other fields, the introduction of partition and fugacity theory to explain environmental distribution of chemicals has occurred. In addition, the use of properties measured in the laboratory to assess behavior and effects in the environment has allowed the development of a clear understanding of many environmental processes. Theoretical methods to predict environmental properties of chemicals have been placed on a sound footing and provide a basis for expansion of knowledge in the future.

The global problem of the **greenhouse effect** now occupies the center stage of environmental management. The environmental chemistry of the processes leading to this problem provides the starting point in its resolution. A range of different scientists have pointed out the importance of developing a mechanistic chemical model of the cycles of carbon and other elements and their interaction with the land and sea in devising strategies for management of this global problem.

1.4 THE SCOPE OF ENVIRONMENTAL CHEMISTRY

It is difficult to precisely define environmental chemistry since the topic has not yet reached a stage where there is universal accord in the chemical community on its scope. However, this is developing and the following definition provides a reasonably acceptable statement at this stage:

Environmental chemistry is the study of sources, reactions, transport, and fate of chemical entities in the air, water, and soil environments, as well as their effects on human health and the natural environment.

Environmental chemistry is probably the most interdisciplinary of the many branches of chemistry. It contains aspects of related branches of chemistry, such as organic chemistry, analytical chemistry, physical chemistry, and inorganic chemistry, as well as more diverse areas, such as biology, toxicology, biochemistry, public health, and epidemiology. Figure 1.1 represents a diagrammatic illustration of some of the major aspects of these relationships. To place the area in some perspective, a set of topics that fall wholly or partially within environmental chemistry are listed in Table 1.2. Many of these topics are concerned with chemical pollutants in the environment,

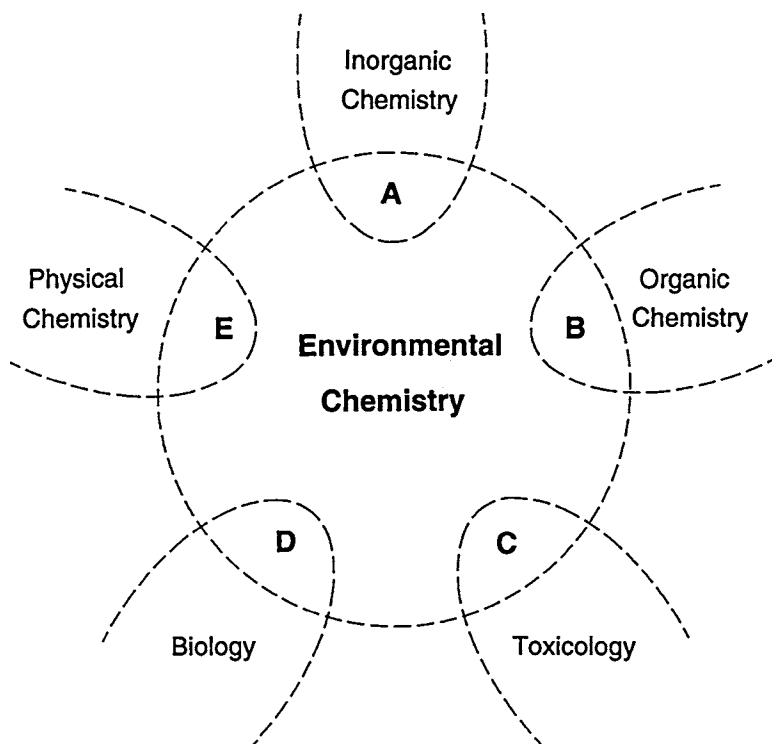


FIGURE 1.1 Diagrammatic illustration of some of the relationships between environmental chemistry and other areas of science.

but environmental chemistry is not only concerned with pollution, but also with the behavior of natural chemicals in natural systems. This is exemplified by topics such as oceanic and limnological chemistry, which are primarily concerned with natural systems.

Investigations within the scope of environmental chemistry provide a possible explanation for our very existence. The primitive Earth's atmosphere contained simple gases that, on equilibration, followed by subsequent complex reaction sequences, have led to the formation of proteins, carbohydrates, fats, and other substances that are the basic molecules needed for life to develop. Later processes in this sequence may have led to the formation of cells and then on to more complex life-forms. The mechanisms of these processes draw heavily on our present understanding of the fundamental properties of molecules and reaction mechanisms.

Environmental chemistry is basically concerned with developing an understanding of the chemistry of the world in which we live. Such investigations and knowledge have an intrinsic value of their own. Our world, with all of its many complex chemical processes, is a worthy topic for a well-rounded education and research. The chemical processes in soil, water, and the atmosphere are central to our existence, and through them we can better understand ourselves and our role in the environment. Thus, it could be argued that environmental chemistry should be a part

TABLE 1.2
Some Topics Considered to Be Wholly or Partially within Environmental Chemistry

Aspect of the Environment	Area within or Relevant to Environmental Chemistry
Evolution	Chemical evolution
Chemical processes in sectors of the abiotic natural environment	Oceanic, atmospheric, soil, and limnological chemistry, global chemical systems
Chemical influences in natural ecosystems	Chemical ecology, pheromones, allelochemistry
Behavior of hazardous chemicals in the environment	Mathematical modeling of environmental distribution, degradation processes, waste disposal
Effects of toxic chemicals on individuals, populations, and ecosystems	Environmental toxicology, ecotoxicology, quantitative structure-activity relationships (QSARs), environmental analysis
Effects of chemicals on human populations	Environmental health, safety, occupational health, epidemiology

of the chemistry curriculum wherever chemistry is taught. At a tertiary level, graduates in chemistry are in fact now expected to be familiar with many aspects of environmental chemistry. They are being asked by management in industry and within government to advise on the environmental and health effects of chemicals, management and safety of chemicals, and so on.

1.5 CHEMISTRY IN ENVIRONMENTAL MANAGEMENT

There is little doubt that the principal applications of environmental chemistry are in the development of an understanding of the behavior and effects of discharged chemicals on human health and the natural environment. When chemical use results in environmental contamination, it is necessary to set standards for acceptable concentrations in water, air, soil, and biota. Monitoring of these concentrations, and the resultant effects, must then be undertaken to ensure that the discharge standards are realistic and provide protection from adverse effects. Also, considerable attention is now being focused on regulation of the use of new chemicals and prevention of chemicals that may have adverse effects from entering the marketplace and environment. Most of these actions are undertaken by government and industry, for example, the disposal of chemical wastes generated in highly concentrated form, as well as occurring as trace contaminants in discharges. These chemical management issues involve political, social, and economic problems as well as technical problems, and many new approaches to management will be needed. On the other hand, new industrial processes that generate less chemical waste will be required. Some of the activities undertaken in government agencies and industry, within the scope of environmental chemistry, are outlined in Table 1.3. In addition, there is a considerable

TABLE 1.3
Some Activities of Government and Industry Related to Environmental Chemistry

Objective	Some Actions Taken
Management of industrial emissions	Setting emission standards, monitoring ambient concentrations, disposal of waste, modeling distribution of chemicals
Protection of workers' health	Biochemical and physiological testing, epidemiology, monitoring ambient concentrations, evaluation of adverse effects
Protection of the natural environment	Monitoring contaminants in water, air, soil, and biota, evaluation of adverse effects
Testing and evaluation of new chemicals	Modeling of potential distributions, testing toxicity and other effects

volume of research in environmental chemistry being undertaken. A number of well-established research institutions specializing in environmental chemistry are operating in countries throughout the world.

Environmental chemistry activities in government, industry, and education are growing, to some extent in concert with the concerns expressed regarding chemicals in the environment by the community. At present, despite the enormous benefits that accrue from the many uses of chemicals, the community often sees chemicals as having a negative impact. A negative impact has occurred in many situations, but many of these have been eliminated and increased knowledge has provided a basis for enlightened future management and control. The further development of environmental chemistry will provide access to chemical products that will enhance the lives of people without resulting in detrimental effects. In this way, environmental chemistry will have a significant effect on the future of chemistry, chemists, and the chemical industry, as well as the community in general.

1.6 BASIC CONCEPT OF THIS BOOK

In the early 1970s, a new approach to explaining the environmental effects of chemicals was initiated. Hamelink and coworkers (1971) proposed that the bioaccumulation of organic compounds by biota was governed by the properties of the chemical rather than ecological factors, as was previously thought. Since that time, this concept has been extended considerably, and now it is clear that the distribution of chemicals in the environment relates to the properties of the chemical as well as characteristics of the environment (see Mackay et al., 1992).

The relationship of biological effects, particularly toxicity, to properties of a chemical, such as the olive oil–water partition coefficient, was well established by Ernest Overton in the early 1900s and extended by Covin Hansch during a later period. These concepts were then applied, with considerable success, to explain

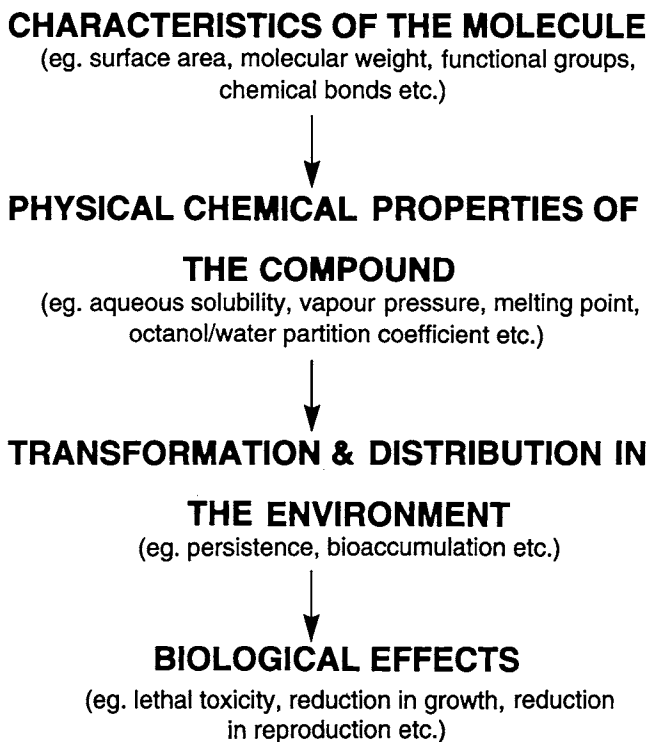


FIGURE 1.2 The environmental chemistry of a chemical can be seen as an interrelated set of characteristics, as shown.

toxicity and other effects of chemicals in the environment. In addition, the influence of characteristics of the molecule itself, such as molecular surface area, was found to be of considerable significance.

A concept of an interdependent set of properties of a chemical has now developed, as shown in Figure 1.2. This has been used as the basic underlying concept of this book. The basic characteristics of the molecule itself are the starting point for understanding the environmental chemistry of chemicals. These characteristics govern the physical-chemical properties of the compound, such as its aqueous solubility and vapor pressure, which in turn control transformation and distribution in the environment. Biological effects relate to the chemical in its transformed and dispersed state. This set of interrelationships also provides a basis for predicting the environmental properties of a chemical. The distribution, transformation, and some biological properties can be predicted in many situations utilizing physical-chemical properties.

This concept provides a framework into which aspects of the environmental characteristics of a chemical can be logically placed. It allows these many diverse environmental characteristics to be rationalized into a set of relationships, rather than be seen as a disorganized collection of facts.

1.7 KEY POINTS

1. The alchemists were the earliest practitioners of applied chemistry, but alchemy was more of an art than a science. Many alchemists were concerned with mysticism and the transmutation of base metals such as lead into gold.
2. Environmental chemistry had an early beginning with Antoine Lavoisier, a French scientist born in Paris in 1743. Lavoisier combined classical experiments on the composition of air and its use by animals to investigate the chemical nature of the atmosphere.
3. The commercial application of chemical knowledge derived from laboratory experiments was commenced by William Perkin in the late 1800s. Perkin founded the synthetic dye industry, which was the forerunner of the large and diverse synthetic chemical industry of today.
4. Perhaps one of the most important environmental applications of chemical knowledge has been the development, by Fritz Haber in the early 1900s, of a commercial process to combine atmospheric nitrogen with hydrogen to produce ammonia. This allowed the manufacture of artificial fertilizers to enhance food crop production and the expansion of the human population.
5. The term *environmental chemistry* has become common only in recent times, although environmental aspects of chemistry have been investigated since the beginning of chemistry itself.
6. Environmental chemistry is the study of the sources, reactions, transport, and fate of chemical entities in air, water, and soil environments, as well as their effects on human health and the natural environment.
7. Over the last 15 years, theoretical concepts of the distribution, behavior, and effects of chemicals in the environment have been developed that provide a fundamental understanding of these processes.
8. Environmental chemistry plays a major role in government and industry in relation to many areas, including management of industrial emissions, protection of workers' health, protection of the natural environment, and testing and evaluation of new chemicals.
9. The environmental properties of a chemical can be seen as an interrelated set of properties based on the characteristics of the molecule. These characteristics govern the physical-chemical properties of the compound, which in turn control transformation, distribution, and biological effects in the environment.

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QUESTIONS

1. The terms *Stone Age*, *Bronze Age*, *Industrial Age*, and *Atomic Age* are used to describe phases of knowledge and technology that human civilizations have passed through. Why is the term *Chemical Age* appropriate for today's society?
2. The applications of chemistry in industry have had a major effect on human society. Which chemical applications have had the greatest impact on the human and natural environment?
3. Environmental chemistry has developed from a diverse field of investigations conducted in relation to the environment. Which of these areas of investigation have been the most important in influencing the scope of environmental chemistry as it is perceived today?

ANSWERS

1. The chemical industry is one of the largest industries in our society.
 - Chemical products dominate the home, workplace, agriculture, transport, medicine, and other areas of human activity.
 - Waste chemical control is now a major aspect of environmental management for industry and government.
2. The artificial fixation of nitrogen to produce ammonia that is used as an agricultural fertilizer has allowed the expansion of the human population.
 - Commercial synthetic chemical operations produce products that are used in almost all human activities.
3. Development of analytical techniques for trace organic compounds and other substances has revealed widespread trace contamination.
 - Attribution of adverse biological effects to contamination by trace chemicals.

Part II

Basic Properties of Chemicals in the Environment

2 Bonds and Molecules: Their Influence on Physical-Chemical Properties in the Environment

2.1 INTRODUCTION

The behavior of chemicals in the environment is governed by their physical-chemical properties as well as transformation and degradation processes, which are discussed in Chapter 3. The physical-chemical properties of compounds include such characteristics as boiling point (bp), melting point (mp), solubility in water, and similar properties. These properties are, in fact, measurements made in the laboratory of environmentally relevant characteristics. For example, compounds with low boiling points evaporate rapidly into the atmosphere, whereas compounds that are highly soluble in water disperse readily in streams and rivers. Thus, an understanding of these properties would be expected to give a clearer perception of how compounds will behave in the environment. This means that the measurement of the physical-chemical properties of a compound could be used to provide an evaluation of its environmental distribution in air, water, sediments, soil, and animals.

2.2 STATES OF MATTER IN THE ENVIRONMENT

A cursory examination of the nature of our environment reveals that matter exists in basically three states: solid, liquid, and gas. Solids are present as soil, rocks, and so on, whereas liquids are represented by the great water bodies of the oceans, lakes, and rivers. The gaseous component of the environment is the atmosphere. These states exercise a basic influence on the nature and distribution of all substances, both natural and man-made, in the environment in basic chemical terms. **Solids** have a definite volume and shape and consist of large numbers of particles that could be atoms, ions, or molecules (Figure 2.1). In fundamental terms, solids are matter where attractive forces between the particles present are more powerful than disruptive forces, and these hold the particles in a fixed relationship to one another. This also means that the particles are closely packed together, with the consequent effect of making solids relatively dense. Some solids can be converted into liquids by heating.

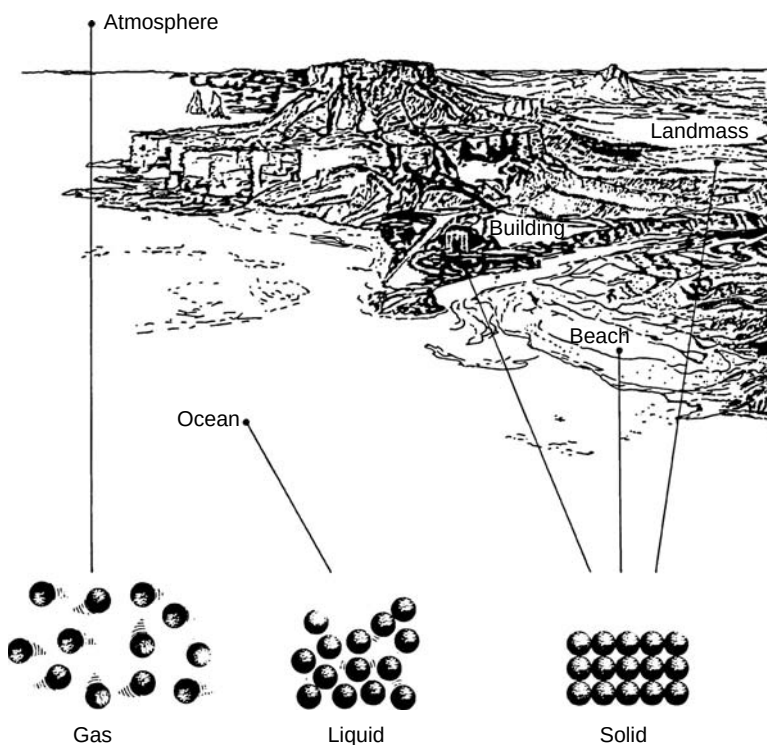


FIGURE 2.1 Illustration of the different states of matter in the natural environment.

For example, ice melts to form liquid water. Heating actually increases the vibrational motions of the particles within the solids until the particles can no longer be held in a fixed relationship to one another. At this point, the substance is converted into a liquid, and the temperature at which this occurs is referred to as the melting point (mp). For example, the melting point of ice at atmospheric pressure is 0°C .

Liquids have a definite volume but no definite shape. With liquids, the particles are close together, almost as close as solids, but are free to move relative to one another (Figure 2.1). This means that the cohesive forces holding the particles together are balanced by the dispersive forces. So the waves of the oceans and the flow of water in rivers represent the basic properties of liquids. An important characteristic of many liquids is their boiling point (bp), which is the temperature at which the liquid vaporizes to give a gas. For example, water has a boiling point of 100°C at atmospheric pressure. However, it should be kept in mind that water does not have to boil in order to vaporize. When the temperature is increased and the rate of vibration of the particles in a liquid increases, some particles bounce completely out of the liquid before the boiling point is reached. In this manner, liquids (and to a lesser extent solids) exhibit a pressure above the surface, referred to as the vapor pressure (vp). Water evaporates from the surface of the oceans, lakes, and rivers without reaching its boiling point. As the temperature increases, the rate of evaporation increases. Somewhat similarly, gases from the atmosphere may

dissolve to some extent in water bodies, although the water bodies are at much higher temperatures than the boiling point of the gas. Thus, oxygen and nitrogen gases are present in low concentrations in seawater.

Gases and vapors are produced when the thermal energy input into the liquid reaches a level whereby the movements of the particles (kinetic energy) become greater than the cohesive energy. The particles move apart at this temperature, which is the boiling point, and move at relatively high velocity, colliding with one another and other substances present in the atmosphere. In this way, the molecules in the atmosphere are in constant motion and collide with one another and with the walls of any container, leading to the phenomenon known as pressure. At sea level, the pressure is about 1 atmosphere (atm), and this pressure falls with distance away from the Earth's surface. As both the density of the atmosphere and the pressure fall, the distance between the molecules in the atmosphere increases and the mean free path between collisions increases. So, at sea level, a molecule has a mean free path of about 10^{-6} cm. At a height from the Earth's surface of 100 km, the pressure has fallen to 3×10^{-7} atm, with a molecule having a mean free path of about 10.0 cm. At a height of 500 km, the mean free path has increased markedly, now exceeding 2×10^6 cm (20 km), while the pressure is $<10^{-6}$ atm.

The properties of some substances in the environment are shown in Table 2.1. Oxygen and nitrogen have relatively low bp and mp values and are the major gaseous components in the atmosphere. Water has higher bp and mp values and exists as a liquid, with the molecules in close association but free to move in relation to one another. Quartz and common salt have very high bp and mp values, reflecting the close and fixed relationship the atoms present have to one another.

2.3 NATURE OF BONDS

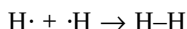
To understand why matter exists as solids, liquids, or gases and how they physically evaporate, dissolve, and generally distribute in the environment, we must start by considering the molecular nature of chemical compounds and the way atoms are bound to one another. First, we will look at the nature of chemical bonds, and this

TABLE 2.1
Properties of Some Substances in the Environment

Substances	Occurrence in the Environment	Normal Physical State	Boiling Point (°C) (Atmospheric Pressure)	Melting Point (°C)
Oxygen	21% of atmosphere	Gas	-183	-218
Nitrogen	78% of atmosphere	Gas	-196	-210
Water	Oceans, lakes, rivers	Liquid	100	0
Common salt	3.5% of seawater	Solid	1413	801
Quartz	Rocks, sand, geological strata	Solid	2230	1610

will give an insight into physical-chemical properties that will provide an understanding of environmental properties.

The most important type of bonding is the covalent bond. Usually with this type of bonding, two atoms react together, with each contributing one electron to form a bond. Thus, two atoms of hydrogen can react to form a hydrogen molecule with one covalent bond. Thus,



Each covalent bond consists of two electrons moving rapidly between the hydrogen atoms in a defined space, as illustrated in Figure 2.2. Each electron holds a full negative charge, and if the electrons spend equivalent times near the two atoms in the bond, this results in no difference in charge between the two ends of the bond. This can be interpreted as the electron density around the two atoms being symmetrical, leading to a nonpolar bond. However, a different situation applies with the hydrogen chloride bond. In this bond, the chlorine tends to attract electrons as shown in Figure 2.3. This attraction is not sufficient to cause the electron to remain permanently with the chlorine atom, but causes the electron to spend more of its time when it is moving between the two atoms toward the chlorine end of the bond. This results in a small partial negative charge (denoted as δ^-) occurring on the chlorine atom and leads to a small partial positive charge (denoted as δ^+) occurring at the hydrogen atom. These factors are illustrated with the hydrogen chloride bond in Figure 2.3. The outcome of this effect is that the bond becomes **polar** and has a **dipole moment** or **polarity**. Quantitatively, the polarity of bonds can be characterized by two factors: (1) the charge on each atom and (2) the distance between the two atoms.

The combination of these two factors allows the development of a quantitative characteristic described as the **dipole moment**, which is defined as follows:

$$\begin{aligned} \text{Dipole moment (DM)} &= \text{in Debye units (D)} \\ &\text{Electrostatic charge} \times \text{Distance between atoms} \end{aligned}$$

If both atoms, or ends of the bond, are the same, as with the hydrogen (H_2) and chlorine (Cl_2) molecules, then there can be no differences in the overall electron density in the bond and the dipole moment for such symmetrical bonds must be zero. However, most unsymmetrical bonds have a dipole moment. For example, the hydrogen chloride bond has a dipole moment of 1.03 Debyes (D); there are examples of the dipole moments of bonds contained in Table 2.2.

It can be expected that a bond will be polar if the two atoms involved differ in their ability to attract electrons. This property is described as the **electronegativity** of an atom and can be estimated quantitatively. The order of magnitude of the electronegativity of some common elements is shown below:



Hydrogen molecule

electron density
(diagrammatic)

Chlorine molecule

electron density
(diagrammatic)

Points:

- * the electron density is symmetrical
- * there is no polarity
- * the bond is nonpolar

FIGURE 2.2 Illustration of the characteristics of nonpolar covalent bonds in the hydrogen and chlorine molecules.

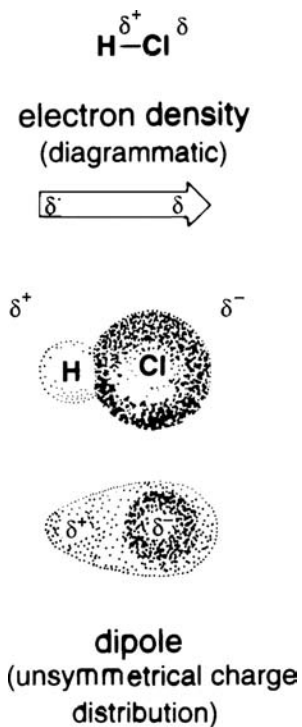


FIGURE 2.3 Illustration of the characteristics of polar covalent bonds using the hydrogen chloride molecule H-Cl as an example.

TABLE 2.2
Dipole Moments of Some Chemical Bonds

Bond	Dipole Moment (Debye, D)	Bond Description
H-F	1.91	Highly polar
H-Cl	1.08	Polar
H-Br	0.80	Polar
H-O	1.5	Polar
H-N	1.3	Polar
C-H	0.4	Weakly polar
C-Cl	0.5	Weakly polar
C-Br	0.4	Weakly polar
C-O	0.7	Weakly polar
C-N	0.2	Nonpolar

Quantitative measures of the electronegativities of various elements are shown in Table 2.3. There is an approximate correlation between the differences in the electronegativity of two atoms in a bond and its dipole moment. For example, the difference in electronegativity of H and F is 1.9, and the dipole moment is 1.91 D; the H and O difference is 1.4, and DM is 1.5; and the C and O difference is 1.0, and DM is 0.7. The partial negative charge resides at the F and O atoms in these bonds.

TABLE 2.3
Electronegativities of Some Elements

Element	Electronegativity	Element	Electronegativity
Li	1.0	C	2.5
Na	0.9	Pb	1.9
K	0.8	N	3.0
Cr	1.6	P	2.1
Fe	1.8	O	3.5
Cu	1.9	S	2.5
Ag	1.9	F	4.0
Au	2.4	Cl	3.0
Zn	1.6	Br	2.8
Cd	1.7	I	2.5
H	2.1		

2.4 POLARITY OF MOLECULES

The C–H bond is one of the most common in organic compounds. By looking at Table 2.3, it can be seen that the electronegativity difference between these two atoms is 0.4. On the other hand, if we look at the O and H atoms, we find that the electronegativity difference is 1.4. This leads to a description of the C–H bond as being **weakly polar** and the O–H bond as being **polar**. By this means we can provide descriptive terms for the polarity of the various bonds, as shown in Table 2.2.

If we join sets of bonds and atoms together to form covalent compounds, this results in the overall polarity of the compound being due to a combination of the dipole moments of the bonds present. The overall dipole moment for the molecule is due to the spatial resolution or vector sum of the various dipole moments of the bonds present in terms of their size and direction.

Figure 2.4 contains a diagrammatic representation of the methane molecule. Since the hydrogen atoms in this molecule are directed symmetrically toward the corners of a tetrahedron, the spatial resolution of the dipole moments of the C–H bonds would result in a dipole moment for methane of zero. So, even though the C–H bond is a weakly polar bond, the overall dipole moment is zero. However, if we replace one of the hydrogen atoms in methane with a chlorine atom to give monochloromethane (Figure 2.4), this gives an unsymmetrical molecule that has a dipole moment of 1.87 D. Similarly, the symmetrical compound tetrachloromethane, which is also known as carbon tetrachloride, has a dipole moment of zero since it is a symmetrical molecule, and trichloromethane, also known as chloroform, has a dipole moment of 1.02 D.

In this discussion we have looked at atoms in terms of their electronegativity, but common groupings of atoms and functional groups have consistent properties in terms of their effects on polarity of molecules. The polarity classifications of a variety of common groupings are shown in Table 2.4. It is interesting to note that the alkyl groups containing many C–H bonds are listed as nonpolar, whereas the C–H bond is listed as weakly polar. This is due to the spatial orientation of the C–H bonds present, which tends to reduce polarity. Thus, the n-alkanes such as hexane are nonpolar.

A variety of important chemicals widely distributed in the environment are weakly polar to nonpolar. For example, DDT with the structure shown in Figure 2.4 is weakly polar to nonpolar since the weakly polar C–Cl bonds are distributed in such a manner as to reduce the overall polarity of the molecule to a very low level. The polychlorinated biphenyls (PCBs) and most of the chlorohydrocarbon pesticides are also weakly polar to nonpolar. On the other hand, the herbicide 2,4-dichlorophenoxyacetic acid (2,4-D) (Figure 2.4) is a polar molecule because the carboxylic group (–COOH) is polar (Table 2.4).

One of the most important compounds in the environment is water since it is the major component of oceans, lakes, and rivers, and thus has a major impact on the distribution of chemicals. Water has a molecular structure that consists of two hydrogen atoms attached to an oxygen atom, with both O–H bonds residing in a plane at an angle of 104° to each other. Spatial analysis of the direction and strength of these bonds gives a resultant dipole moment of 1.80 D, which is in agreement

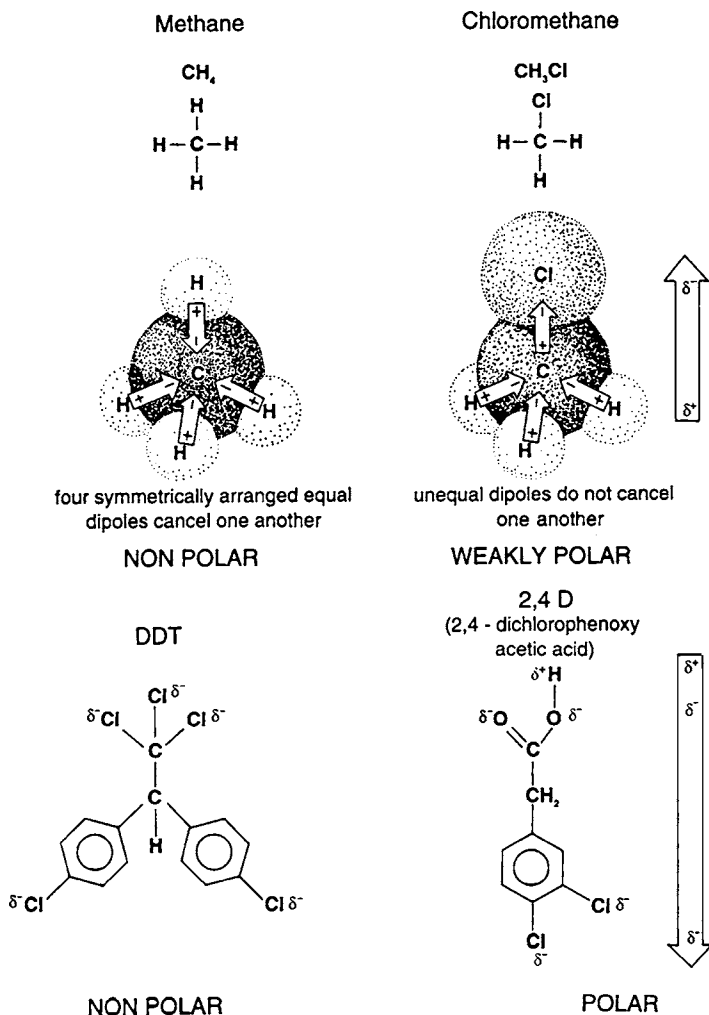


FIGURE 2.4 Chemical structures of some compounds with different polarities.

with the experimental value of 1.87 D (Figure 2.5). As a result, water is a polar molecule with physical properties, and its ability to dissolve other substances is strongly influenced by its polarity.

2.5 IONIC COMPOUNDS

With some elements, the difference in electronegativities can be so great that one atom completely loses an electron to the other atom in the bond. For example, in the hydrogen chloride molecule, there can be a transfer of an electron from the hydrogen to the chlorine, resulting in the formation of a chloride ion with a full

TABLE 2.4
Polarity of Common Groupings in Organic Molecules

Highly Polar	Polar	Weakly Polar	Nonpolar
--COO^-	--O--H	>C--Cl	--CH_3
--O^-	--COOH	>C--Br	$\text{--CH}_2\text{CH}_3$
--NH^{3+}	--NH_2	>C--H	
	>C=O		

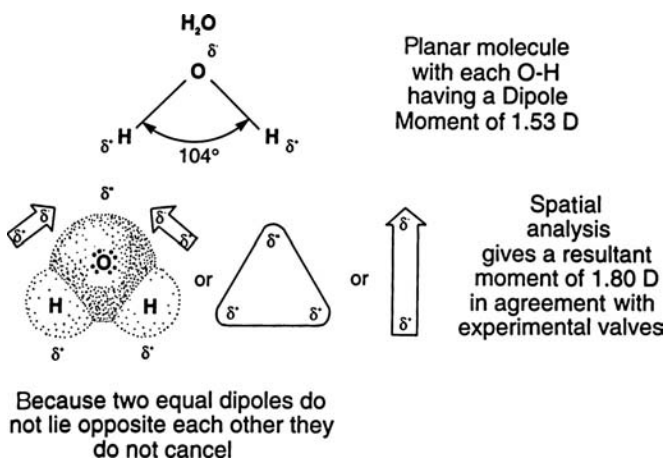
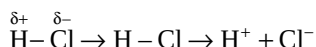


FIGURE 2.5 Representations of the water molecule and its polar characteristics.

negative charge and, consequently, a hydrogen ion with a full positive charge. This is illustrated below:



In this transfer, the arrow indicates that both electrons that form the covalent bond have moved to the chlorine atom. In this situation, the chlorine atom in the HCl molecule, which held a small negative charge (δ^-), now becomes a fully negatively charged free atom, described as an ion. The covalent bond has been broken and no

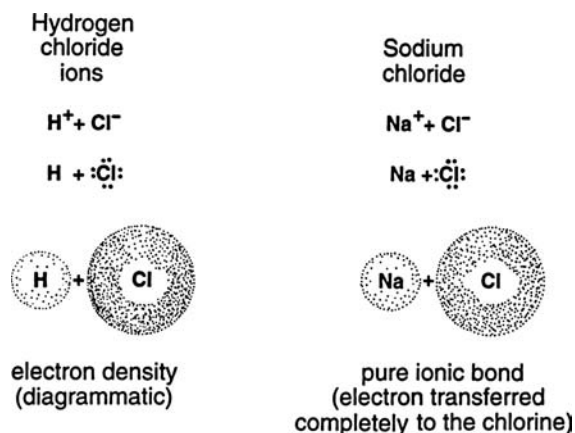
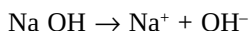


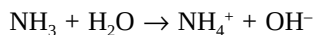
FIGURE 2.6 Formation of ions with hydrogen chloride and sodium chloride.

longer exists (Figure 2.6). With the formation of ions, the hydrogen ion and chloride ion can move apart depending on conditions. In fact, in hydrogen chloride there is a mixture of polar hydrogen chloride molecules together with hydrogen and chloride ions. Measurements indicate that in pure liquid hydrogen chloride, 17% of the compound is in the ionic form. With metal salts of some substances, there can be 100% of the substance in the ionic form. Examples of these substances are sodium chloride, potassium bromide, and so on. Similar processes occur with these substances as with hydrogen chloride, as illustrated in Figure 2.6, but to a greater extent.

Compounds that form ions in aqueous solutions will conduct electricity and are often referred to as **electrolytes**. Acids are compounds that release hydrogen ions into solution, and bases are compounds that release hydroxide ions (OH^-) into solution. For example,

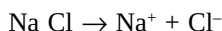
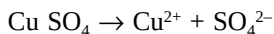


Sodium hydroxide (caustic soda)

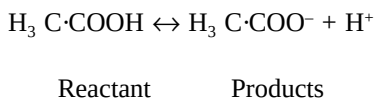


Ammonia

Salts are ionic compounds formed from the cation of a base and the anion of an acid that yield neither a hydrogen ion nor hydroxide ion on dissociation. Some typical salts are



Generally, complete dissociation of salts occurs in water. However, acids may only partially dissociate. For example, strong acids, such as the mineral acids H_2SO_4 (sulfuric) and HNO_3 (nitric), and the organic acids $\text{H}_3\text{C}\cdot\text{COOH}$ (acetic) and $\text{H}\cdot\text{COOH}$ (formic), dissociate only partly. Thus, with acetic acid, an equilibrium is formed that favors the undissociated form



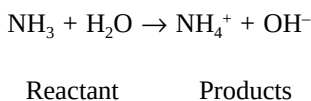
A general equilibrium situation is established where $K = [\text{products}]/[\text{reactants}]$, and in the case of acetic acid,

$$K = \frac{[\text{H}^+][\text{H}_3\text{C}\cdot\text{COO}^-]}{[\text{H}_3\text{C}\cdot\text{COOH}]}$$

where K is the equilibrium constant and $[\text{H}^+][\text{H}_3\text{C}\cdot\text{COO}^-]$ and $[\text{H}_3\text{C}\cdot\text{COOH}]$ are the molar concentrations of the various substances. The value of K can be calculated from actual measurements as

$$K = \frac{(0.00135)(0.00135)}{(0.0986)}$$

K is constant for dilute solutions of acetic acid at 1.85×10^{-5} . This means that the proportion of the compound that dissociates into ions is constant in dilute solution irrespective of the concentration. There is a similar situation with bases. For example, with ammonia an equilibrium reaction occurs in water; thus,



$$K' = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3][\text{H}_2\text{O}]}$$

where K' is the equilibrium constant and $[\text{NH}_4^+]$, $[\text{OH}^-]$, $[\text{NH}_3]$, and $[\text{H}_2\text{O}]$ are the concentrations of the various substances. Since the concentration of water is very high and constant, it is included in K :

$$\therefore K'[\text{H}_2\text{O}] = K = \frac{[\text{NH}_4^+][\text{OH}^-]}{[\text{NH}_3]}$$

From actual measurement, $K = 1.8 \times 10^{-5}$.

With pure water a small proportion of the molecules dissociate and equal concentrations of H^+ and OH^- are formed:



$$K = \frac{[H^+][OH^-]}{[H_2O]}$$

where K is the equilibrium constant and $[H^+]$, $[OH^-]$, and $[H_2O]$ are the molar concentrations of the substances indicated.

Since the undissociated form $[H_2O]$ is present in close to 100% concentration and effectively constant, it is included in K . So,

$$K [H_2O] = K_w = [H^+][OH^-] \text{ and } [H^+] = [OH^-] = \sqrt{K_w}$$

and from actual measurement,

$$K_w = 1 \times 10^{-14} M$$

thus,

$$[H^+] = [OH^-] = 1 \times 10^{-7} M$$

The hydrogen ion concentration present in water is an important characteristic influencing behavior of ions. For convenience, it is placed on a scale that is easy to use; thus,

$$pH = -\log [H^+]$$

This means that pure neutral water has a pH of 7. Since acids release H^+ into the solution, the molar concentrations will be greater than 10^{-7} , i.e., 10^{-3} , 10^{-4} , and so on. This means acid solutions will have $pH < 7$. Alkaline solutions have lower concentrations of H^+ than neutral solutions, i.e., 10^{-8} , 10^{-10} , and so on, and thus have $pH > 7$.

2.6 INTERMOLECULAR FORCES

In the previous section we considered charges that are generated within molecules. It can be appreciated that these charges within molecules will interact with adjacent

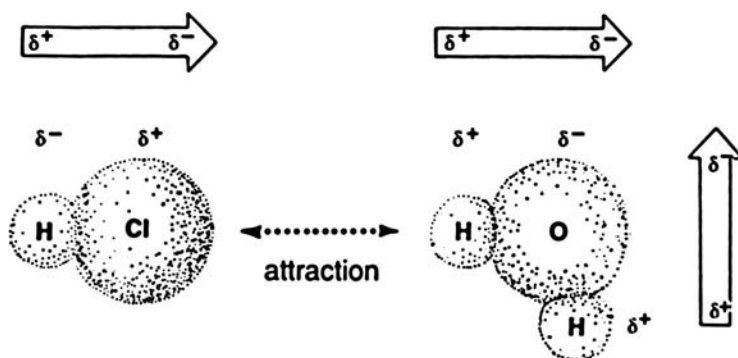


FIGURE 2.7 Attraction between the negative end of a hydrogen chloride molecule and the positive end of a water molecule.

molecules with the usual rules for the interaction of charges applying. This means that like charges result in repulsion and unlike charges result in attraction. Thus, positive-to-positive and negative-to-negative forces result in the generation of a disruptive or repulsive force, and positive-to-negative charges result in an attraction or a cohesive force. Intermolecular forces between covalent compounds are usually described as **van der Waals forces**. Van der Waal was a Dutch scientist who developed an understanding of intermolecular forces. These intermolecular forces have a strong influence on the physical-chemical properties of compounds and how substances behave in the environment.

Molecules that are polar can be attracted to one another so that particular orientations of the positive and negative charges are formed. Examples of these interactions, referred to as **dipole bonding**, are shown in Figure 2.7 and Figure 2.8. Here the δ^- charge on the chlorine in hydrogen chloride is oriented toward the δ^+ on the hydrogen atom in water (Figure 2.7). In water, alcohols, and acids, a network of orientations of the δ^+ hydrogen toward the δ^- of the oxygen in another molecule of the same compound is set up. Of course, in mixtures, these interactions occur between molecules of different compounds. The **hydrogen bond** is a special form of dipole bonding that is of particular significance because it is relatively common and strong. Hydrogen bonding with water, alcohol, and acid molecules is shown diagrammatically in Figure 2.8. Somewhat similarly, polar molecules with dipole moments may interact in various ways with ions.

Another important group of intermolecular interactions are described as **London forces** or **dispersion forces**. These operate over relatively short distances and are responsible for the aggregation of molecules that possess neither free charges nor dipole moments. London forces increase rapidly with molecular weight and are particularly strong with aromatic compounds, but with relatively low molecular weight compounds, these are often relatively weak when compared with dipole interactions.

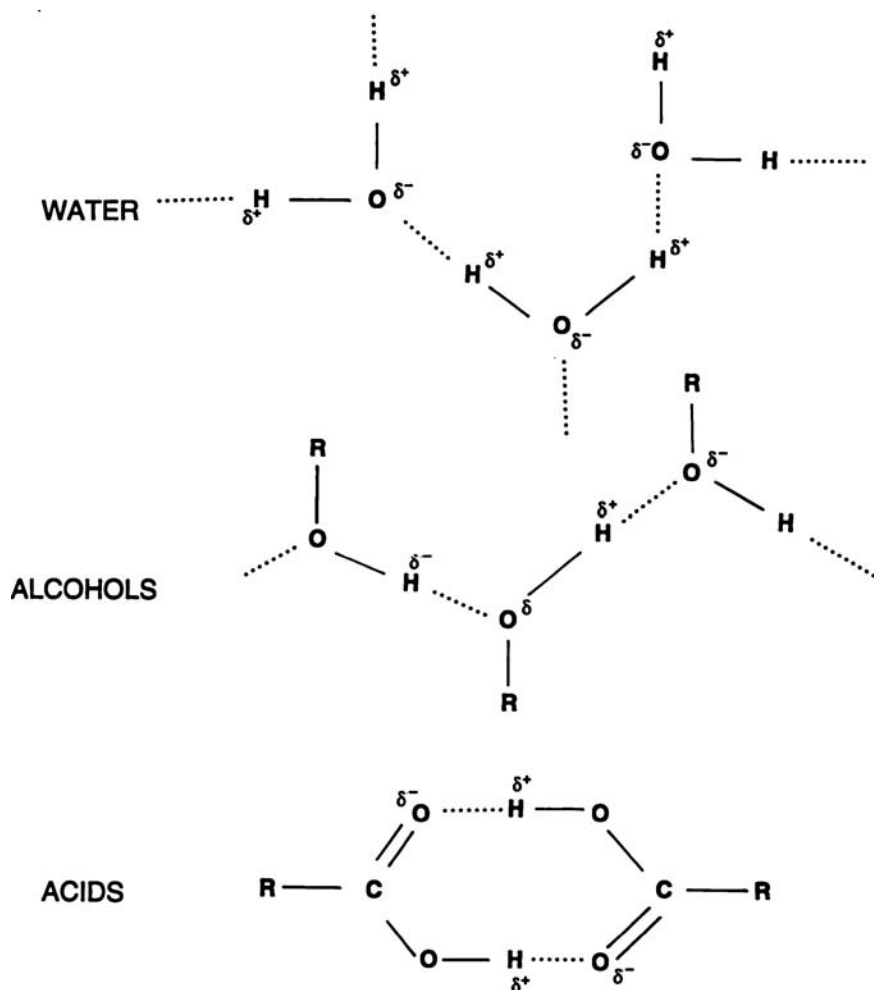


FIGURE 2.8 Hydrogen bonding in water, alcohols, and acids.

2.7 PHYSICAL-CHEMICAL PROPERTIES OF COMPOUNDS

2.7.1 MELTING POINT (MP)

When a substance melts, it changes from a solid to a liquid. In molecular terms, a solid has the component particles in a fixed relationship to one another. There are strong forces between chemical entities as occur, for example, with ions, high-molecular-weight substances, and highly polar substances. With ionic solids, each of the ions is held in a fixed relationship to other ions by the strong forces between the ions. This is illustrated diagrammatically with sodium chloride in Figure 2.9. This compound is held in the solid form by the fixed and close relationship between